BOOK RECOMMENDATIONS POC · PROTOTYPE WALKTHROUGH

Your shop, with recommendations switched on

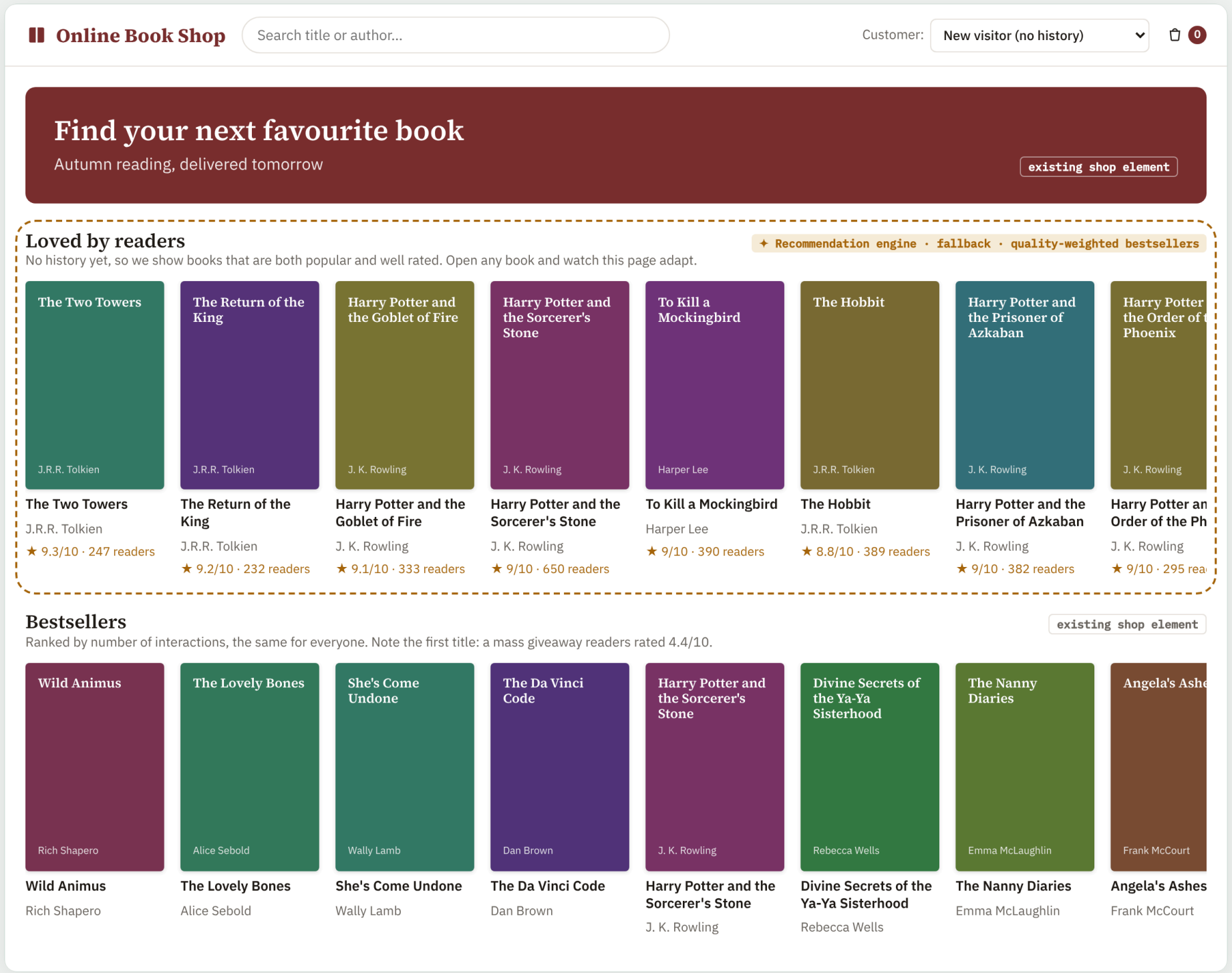
A mock-up of your online book shop running the PoC model (item-to-item similarity, BM25) on your data. Everything marked "✦ Recommendation engine" is new; the rest is what a shop already has.

Live and interactive: **https://dvorpet.github.io/Recommend-book/#prototype**

Note: in these screenshots book covers are drawn placeholders. On the live site real covers load from the image links in your export.

Homepage · new visitor

No history yet, so the engine falls back to books that are both popular and well rated (dashed box). Below it, the shop's existing bestseller list: the same for everyone, led by a mass giveaway rated 4.4/10.



Book page · The Two Towers

“Readers who liked this also enjoyed” is the item-to-item model: each tip shows how many readers read both books. The bestseller box on the right is the shop’s existing element, identical on every page.

Online Book Shop

Search title or author...

Customer: New visitor (no history)

0

Home › J.R.R. Tolkien › The Two Towers



The Two Towers

J.R.R. Tolkien · 1986

★★★★★ 9.3/10 · 247 readers

The Two Towers (The Lord of the Rings, Part 2)

Add to basket

Price, stock and blurb come from your catalogue, which this mock-up does not have.

Bestsellers

`existing shop element`

1. Wild Animus
2. The Lovely Bones
3. She's Come Undone
4. The Da Vinci Code
5. Harry Potter and the Sorcerer's Stone

Same list on every page.

Readers who liked this also enjoyed

Ranked by how strongly the same readers connect the two books

`✦ Recommendation engine · item-to-item`



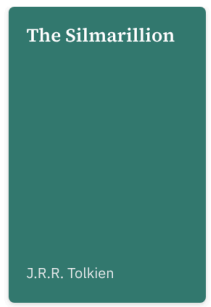
The Return of the King
J.R.R. Tolkien
166 readers read both



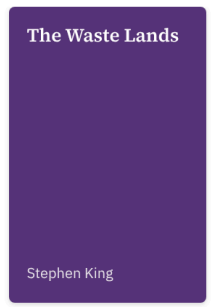
The Fellowship of the Ring
J.R.R. Tolkien
165 readers read both



The Hobbit
J.R.R. Tolkien
111 readers read both



The Silmarillion
J.R.R. Tolkien
33 readers read both



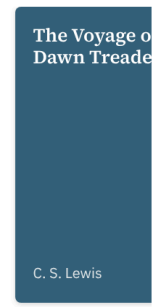
The Waste Lands
Stephen King
23 readers read both



The Tolkien Reader
J. R. R. Tolkien
10 readers read both



Wizard and Glass
Stephen King
25 readers read both

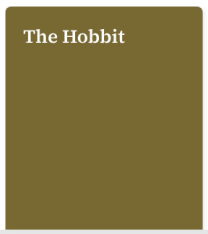


The Voyage of t Dawn Treader
C. S. Lewis
11 readers read b

More from J.R.R. Tolkien

Simple rule that also covers brand-new titles with no readers yet

`✦ Recommendation engine · rule · same author`





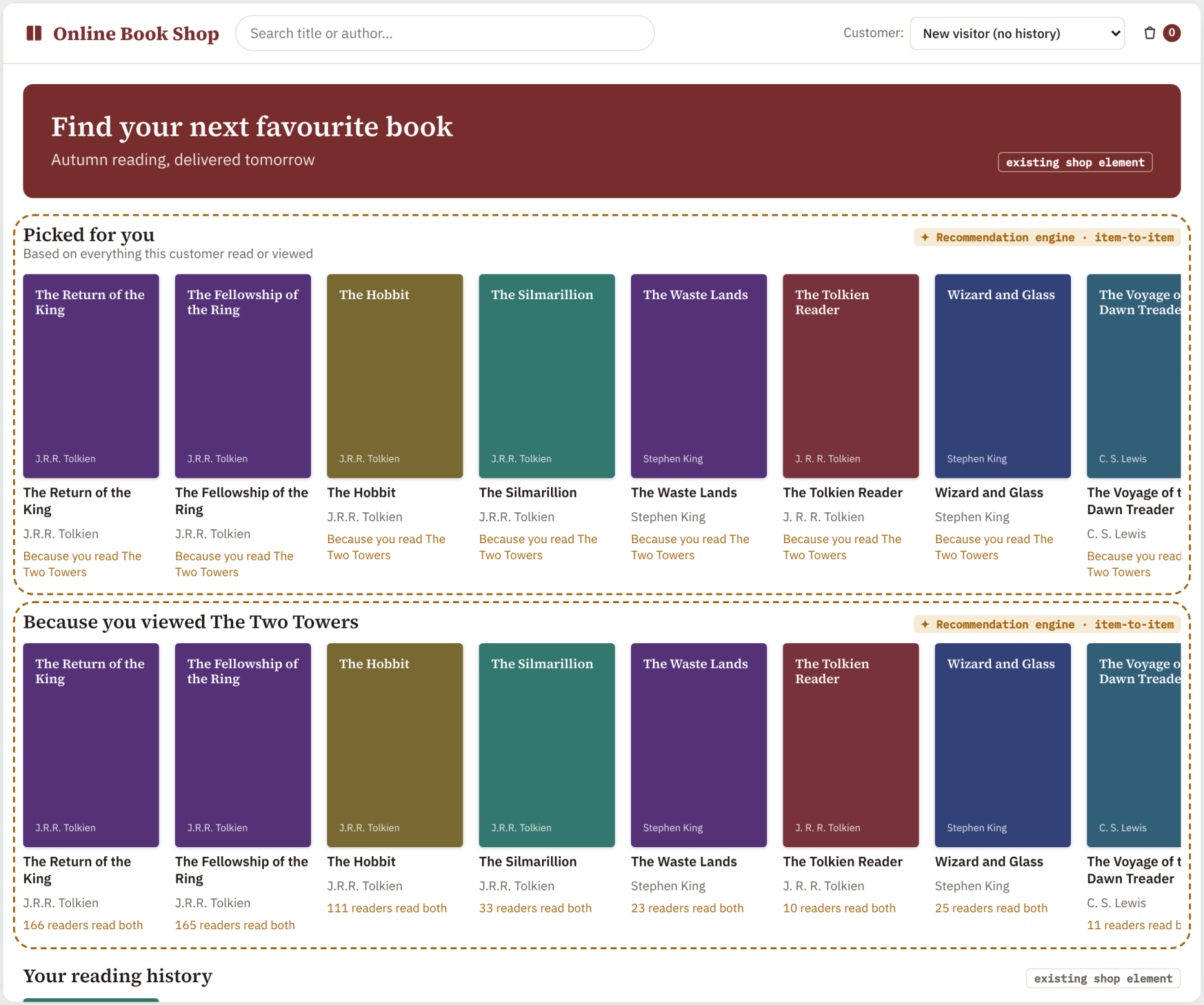






Homepage · after viewing one book

The same visitor after opening one book. The page adapts immediately with “Picked for you” and “Because you viewed …”. This is how the engine handles cold start.



Homepage · fantasy reader

A known customer with three fantasy books in their history gets personal picks, each with the reason (“Because you read …”).

Online Book Shop | Search title or author... | Customer: Fantasy reader | 0

Find your next favourite book

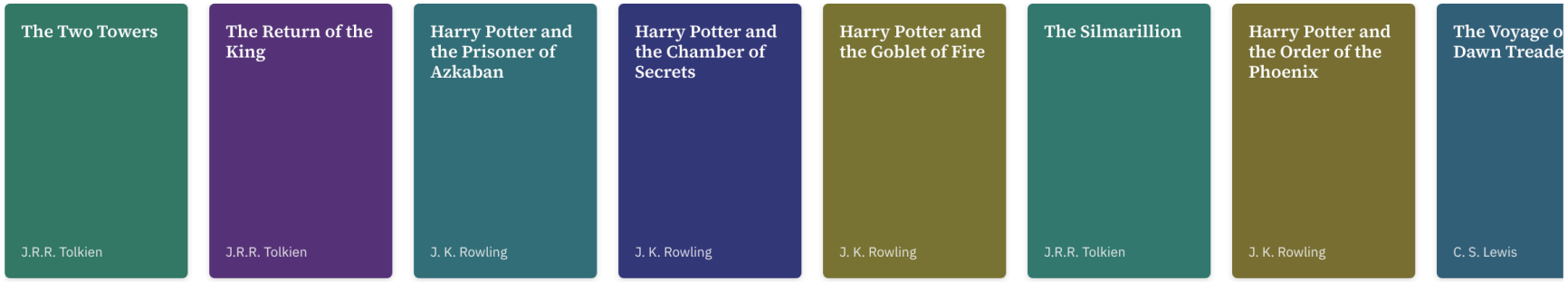
Autumn reading, delivered tomorrow

existing shop element

Picked for you

Based on everything this customer read or viewed

✦ Recommendation engine · item-to-item



The Two Towers
J.R.R. Tolkien
Because you read The Fellowship of the Ring

The Return of the King
J.R.R. Tolkien
Because you read The Fellowship of the Ring

Harry Potter and the Prisoner of Azkaban
J. K. Rowling
Because you read Harry Potter and the Sorcerer's Stone

Harry Potter and the Chamber of Secrets
J. K. Rowling
Because you read Harry Potter and the Sorcerer's Stone

Harry Potter and the Goblet of Fire
J. K. Rowling
Because you read Harry Potter and the Sorcerer's Stone

The Silmarillion
J.R.R. Tolkien
Because you read The Fellowship of the Ring

Harry Potter and the Order of the Phoenix
J. K. Rowling
Because you read Harry Potter and the Sorcerer's Stone

The Voyage of t Dawn Treader
C. S. Lewis
Because you read Fellowship of the

Your reading history

existing shop element



Harry Potter and the Sorcerer's Stone
J. K. Rowling

The Fellowship of the Ring
J.R.R. Tolkien

The Hobbit
J.R.R. Tolkien

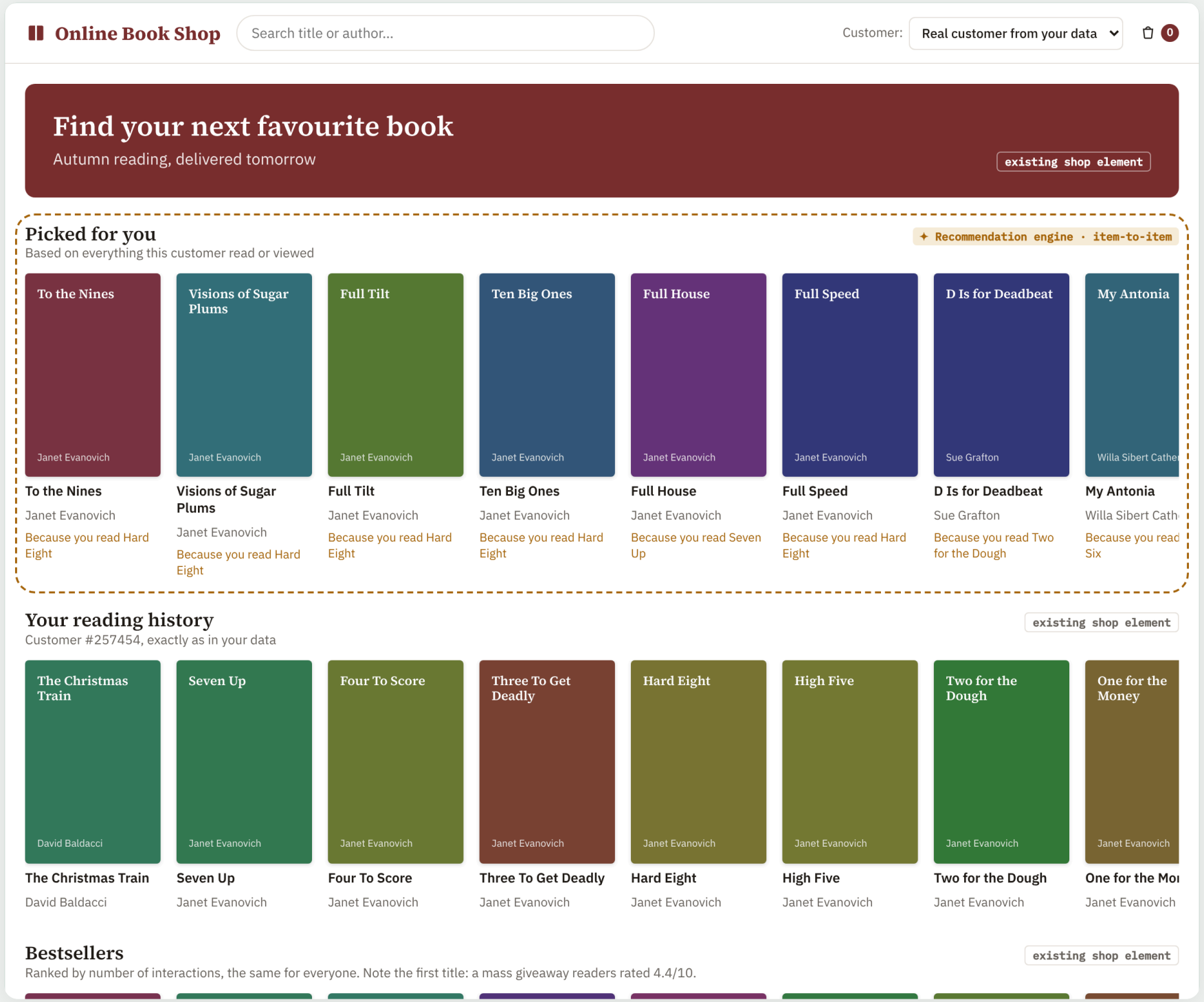
Bestsellers

existing shop element

Ranked by number of interactions, the same for everyone. Note the first title: a mass giveaway readers rated 4.4/10.

Homepage · real customer from the data

Customer #257454 exactly as recorded: a Stephanie Plum fan. The model recommends the rest of the series, then Sue Grafton, a related author.



What the customer sees

The same page with highlighting switched off. The recommendation blocks blend into the shop like any other content.

Online Book Shop

Search title or author...

Customer: Real customer from your data

0

Find your next favourite book

Autumn reading, delivered tomorrow

Picked for you

Based on everything this customer read or viewed



To the Nines
Janet Evanovich
Because you read Hard Eight

Visions of Sugar Plums
Janet Evanovich
Because you read Hard Eight

Full Tilt
Janet Evanovich
Because you read Hard Eight

Ten Big Ones
Janet Evanovich
Because you read Hard Eight

Full House
Janet Evanovich
Because you read Seven Up

Full Speed
Janet Evanovich
Because you read Hard Eight

D Is for Deadbeat
Sue Grafton
Because you read Two for the Dough

My Antonia
Willa Sibert Cath
Because you read Six

Your reading history

Customer #257454, exactly as in your data



The Christmas Train
David Baldacci

Seven Up
Janet Evanovich

Four To Score
Janet Evanovich

Three To Get Deadly
Janet Evanovich

Hard Eight
Janet Evanovich

High Five
Janet Evanovich

Two for the Dough
Janet Evanovich

One for the Mo
Janet Evanovich

Bestsellers

Ranked by number of interactions, the same for everyone. Note the first title: a mass giveaway readers rated 4.4/10.

How it works

Four steps from raw data to recommendations, the offline result against baselines, and what the prototype deliberately is not.

1

Clean the data

Keep ratings with catalogue metadata, merge editions into works, drop bulk-upload accounts and explicit dislikes, keep readers and books with 5+ interactions.

2

Learn which books go together

Item-kNN with BM25 weighting: two books are similar when the same readers read both, discounted for very popular titles so bestsellers do not drown everything.

3

Recommend

Book page: the most similar books. Personal picks: add up similarities from everything in the reader's history, and keep the book that contributed most as the reason. No history: quality-weighted bestsellers.

4

Serve

Here: a static JSON file read by this page. In production: a nightly batch job writes the lists to a fast key-value store behind an API with business rules (appendix A7).

Offline result on this data

MODEL	NDCG@10	HIT RATE@10	COVERAGE
Bestsellers (count)	0.019	8.1 %	0.1 %
ALS	0.054	18.2 %	8.5 %
Item-kNN (BM25)	0.066	19.2 %	76.5 %

What this prototype is not

- Ratings only: no orders, prices, dates or clicks yet
- No stock, price, age or business rules yet
- Covers come from the image links in your export; missing ones are drawn
- It shows the approach is feasible, not that it sells. That is the A/B pilot.